FOREST RESILIENCE ASSESSMENT
Vaia Storm

A Google Earth Engine Workflow for
Municipality-Level Post-Disturbance Vegetation Monitoring

INDEPENDENT GEOSPATIAL ANALYSIS · PORTFOLIO PROJECT

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Executive Summary

This report documents an independent, geospatial workflow for monitoring forest recovery following the Vaia windstorm (27–30 October 2018), which damaged an estimated 42,500 hectares of forest across 494 municipalities in the eastern Italian Alps (Chirici et al., 2019). Using Google Earth Engine, multi-temporal Sentinel-2 imagery (2018–2025) was processed into an NDVI (Normalized Difference Vegetation Index) time series over a 495 km² study area in the Agordino–Cadore district (Province of Belluno), masked to a pre-storm forest baseline derived from Copernicus CORINE Land Cover 2018. The resulting rasters are agregated into municipality-level zonal statistics, producing a single multi-year attribute table suitable for longitudinal comparison. Results show a pronounced vegetation decline immediately after the storm, followed by a summer-NDVI recovery that approaches pre-storm levels by 2025 in most municipalities. Cross-checking this trend against recent satellite basemaps, however shows that forest canopy structure has visibly not returned to its pre-storm state in the most severely affected areas - a discrepancy attributable to a known limitation of NDVI rather than genuine structural recovery. The workflow is fully reproducible using free, open satellite data.

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1. Introduction

This report presents a geospatial analysis exploring the use of multi-temporal satellite remote sensing to assess forest resilience following the Vaia storm. The project originated from a personal interest in Earth observation for ecosystem monitoring and from a direct connection to the study area, visited repeatedly over several years. It is intended as a proof of concept demonstrating a reproducible, cloud-based workflow.

Between 27 and 30 October 2018, the Vaia storm struck north-eastern Italy with sustained winds exceeding 150–200 km/h, felling an estimated 14 million trees and damaging roughly 42,500 hectares of forest across 494 municipalities in Trentino-Alto Adige, Veneto, Friuli-Venezia Giulia and Lombardy (Chirici et al., 2019). Wind-thrown stands subsequently became vulnerable to a prolonged bark-beetle (Ips typographus) outbreak, compounding the original disturbance in several districts over the following years.

Monitoring vegetation recovery after large-scale disturbances of this kind is central to sustainable forest management, helping to target restoration effort and monitoring resources where they are most needed. Satellite remote sensing offers a scalable, repeatable means of tracking these dynamics; this report applies that approach to a subset of the Vaia-affected area, combining optical, land-cover and administrative datasets into a single, auditable pipeline.

2. Objectives

The analysis pursues three specific objectives:

- Quantify and localise vegetation loss and multi-year recovery (2018-2025) using an NDVI time series and critically assess what that single index can - and cannot - tell us about true forest recovery.
- Resolve results at municipality level, comparing recovery trajectories across the individual municipality within the study area.

3. Study Area

The Area of Interest (AOI) covers approximately 495 km² in the Agordino-Cadore district of the Province of Belluno (Veneto, Italy), within the Dolomites, one of the districts most severely affected by the Vaia storm (Chirici et al., 2019). AOI is defined as a bounding rectangle in WGS84 coordinates (Table 1).

Table 1: AOI boundary vertices (WGS84).

Vertex	Longitude (°E)	Latitude (°N)
1	11.90	46.46
2	12.19	46.46
3	12.19	46.26
4	11.90	46.26

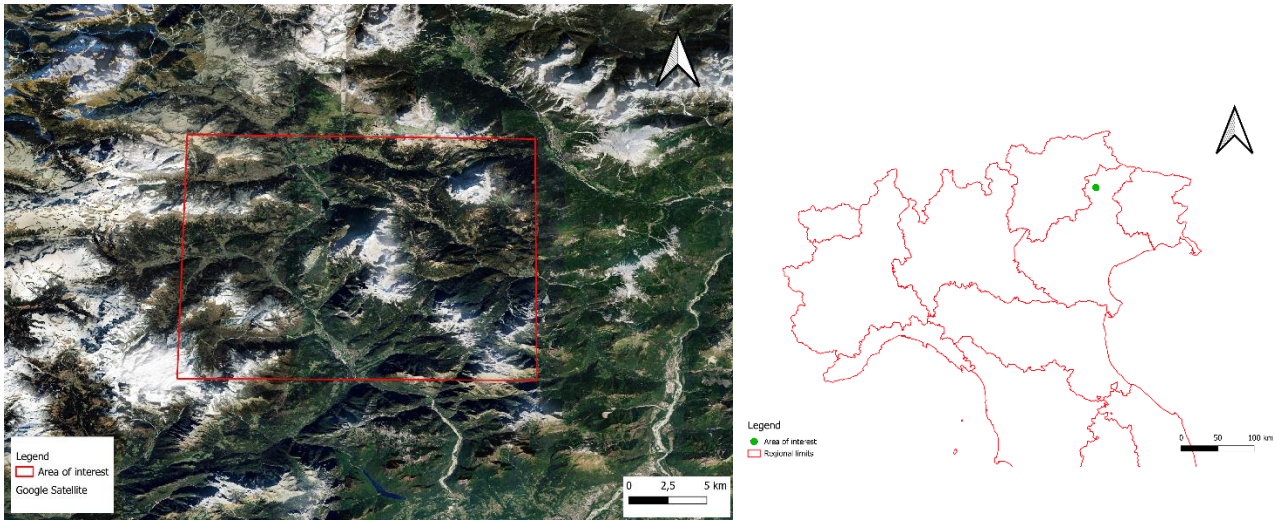


Figure 1: Geographic context of the study area (red rectangle), Agordino–Cadore district, Province of Belluno, Italy.

4. Data & Pre-processing

4.1 Datasets

This analysis combines four categories of input data, all processed within Google Earth Engine's cloud platform, reported in Table 2.

Table 2: Input datasets.

Dataset	Source / Collection	Resolution	Period	Role in workflow
Optical imagery	Sentinel-2 MSI Level-2A (Surface Reflectance)	10 m	2018–2025	NDVI
Cloud mask	Sentinel-2 Scene Classification Layer (SCL)	20 m	2018–2025	Cloud /shadow masking
Forest baseline	Copernicus CORINE Land Cover 2018	100 m	2018 (pre-storm)	Pre-Vaia forest mask
Admin. boundaries	Municipal polygons	vector	current	Zonal statistics units

The number of individual Sentinel-2 scenes contributing to each year's July–August composite (Table 3) varied with satellite revisit frequency and cloud cover; 2019 has noticeably fewer usable scenes than other years.

Table 3: Sentinel-2 scenes used per year (July and August).

Year	2018	2019	2020	2021	2022	2023	2024	2025
Number of scenes	11	7	11	10	10	9	10	10

4.2 Pre-processing

The following pre-processing steps were applied to every Sentinel-2 scenes:

- Cloud and cloud-shadow masking using the Sentinel-2 Scene Classification Layer (SCL): only pixels flagged with the classes below were retained.

Table 4: Sentinel-2 SCL classes retained for cloud masking.

SCL code	Class retained
4	Vegetation
5	Bare soils
6	Water
7	Unclassified
11	Snow / ice

- Filtering of scenes by cloud-cover percentage prior to compositing (threshold: 70%).
- Monthly median compositing (per month, across all available cloud-free scenes) to reduce residual noise; the resulting July and August monthly medians were then averaged into a single annual summer value, suppressing phenological variability between years.

- Masking of all non-forest land cover using a pre-storm (2018) CORINE baseline, the classes considered are reported in Table 5, so that pixels destroyed by Vaia itself are correctly retained in the analysis. The resulting mask is showed in Figure 2.

Table 5: Corine Land Cover 2018 classes retained for the forest mask.

CORINE code	Class retained
311	Broad-leaved forest
312	Coniferous forest
313	Mixed forest

- Clipping of all layers to the Area of Interest.

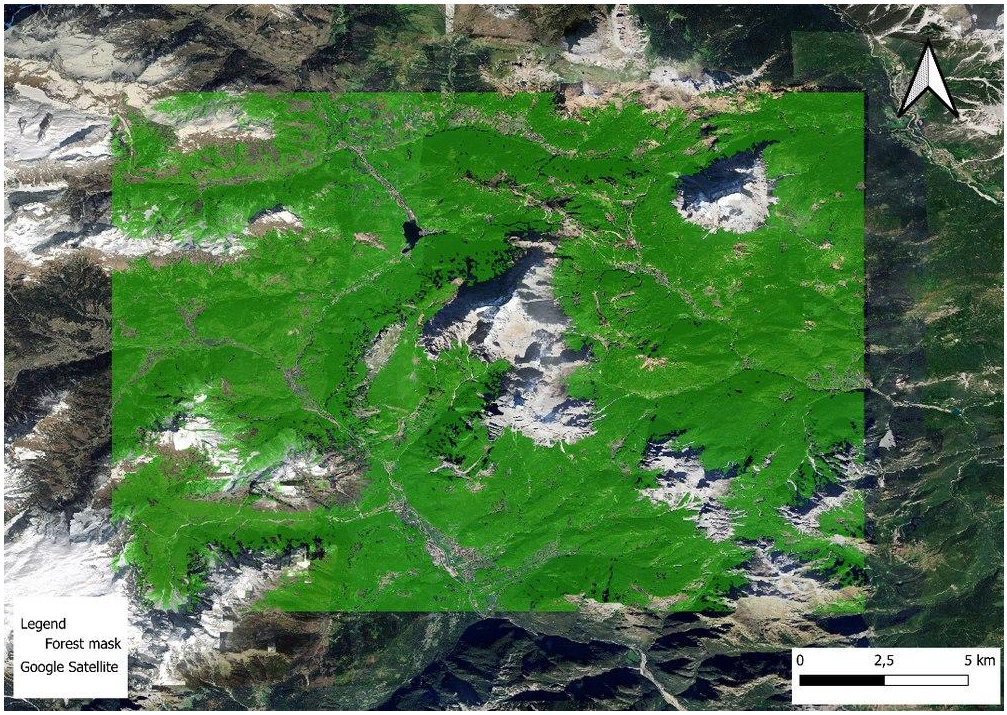


Figure 2: Pre-Vaia forest mask (Copernicus CORINE Land Cover 2018) overlaid on a satellite basemap, clipped to the study area.

5. Methodology

5.1 Vegetation index

A single spectral index, NDVI (Normalized Difference Vegetation Index), was computed from the cloud-masked Sentinel-2 composites to assess overall vegetation greenness and photosynthetic activity:

$$\textbf{NDVI} = (NIR - Red) / (NIR + Red)$$

NDVI was chosen as the entry-point metric for this proof of concept because it is simple, well-understood and requires only two spectral bands. As Section 7 discusses, however NDVI on its own is not sufficient to confirm genuine forest recovery and other complementary indices are identified there as the logical next additions to this workflow.

5.2 Temporal analysis

To minimise seasonal variability, only summer observations (July–August) were retained for inter-annual comparison: each month was first reduced to a median composite and the two monthly medians were then averaged into a single annual summer NDVI value. These annual composites were compared across three phases: pre-disturbance conditions (2017–2018), the immediate post-disturbance response (2019) and medium-term recovery (2020–2025). NDVI differencing between pre- and post-storm composites (ΔNDVI) was used to localise the areas of most severe canopy loss and prioritise them for closer analysis.

5.3 Municipality-level aggregation

For each year, a spatial average of NDVI was computed for every municipality within the AOI to assess the resilience of the area. This produced a shapefile with annual summer values of NDVI for each municipality, enabling temporal comparison of recovery trajectories across the municipalities.

6. Results

The ΔNDVI I (Figure 3) identifies the areas of most severe canopy loss across the AOI. Cross-checking these ΔNDVI derived critical areas against the official Vaia damage perimeters declared by Regione Veneto (Figure 4) shows a close spatial coincidence between the two.

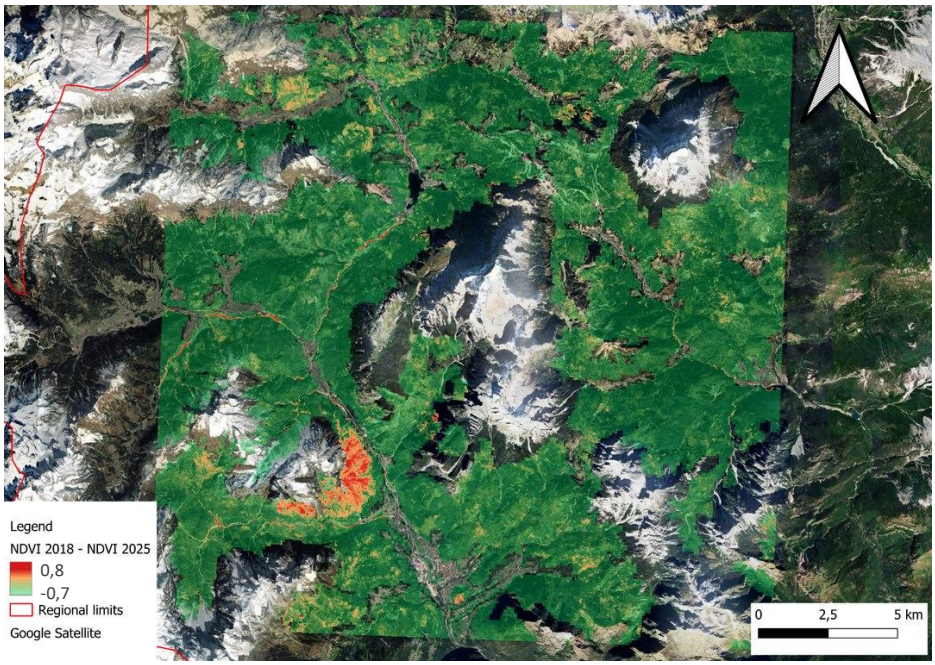


Figure 3: NDVI difference (ΔNDVI) between pre- and post-storm summer composites. Larger values indicate more severe canopy loss and identify priority areas for the recovery analysis.

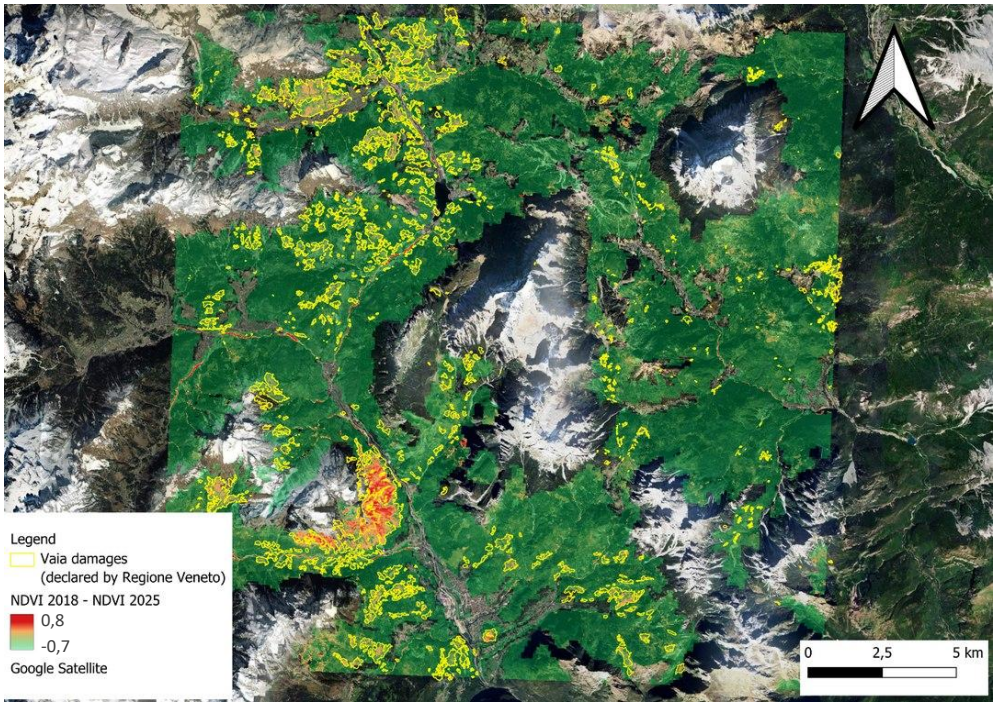


Figure 4: ΔNDVI-derived disturbance extent cross-checked against official Vaia damage perimeters declared by Veneto Region for qualitative validation.

Turning to the temporal trend, Figure 6 shows NDVI trajectories for the 10 municipalities with the greatest loss (number of wind-thrown trees > 100). The areas corresponding to each municipality is clarified in Figure 5. The analysis reveals a clear, storm-attributable decline in NDVI in 2018–2019, followed by a steady multi-year recovery.

- NDVI dropped sharply in the first full growing season after the storm (2019) across all municipalities in the AOI, consistent with the sudden, widespread canopy loss documented for this event (Chirici et al., 2019).
- Recovery from 2020 onward was and by 2025, summer NDVI approaches or exceeds 2018 (pre-storm) levels in most municipalities - which, taken at face value, would suggest the forest has essentially recovered.
- Cross-checking that results against recent high-resolution satellite basemaps tells a different picture: in the most severely affected stands, the forest canopy has visibly not regrown to its pre-storm structure, despite NDVI values suggesting otherwise.

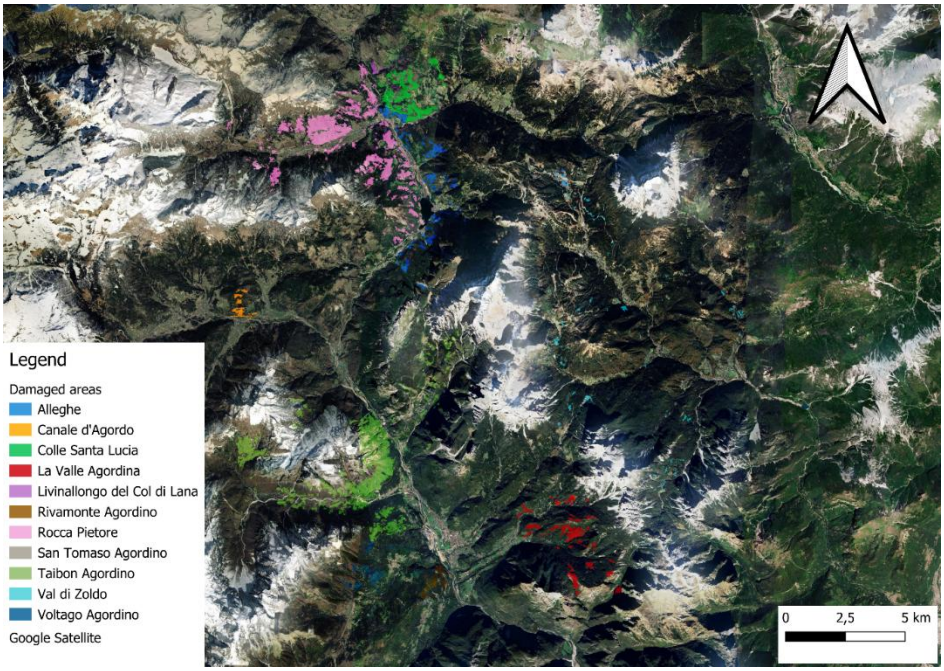


Figure 5: Damaged areas coloured according to their municipality.

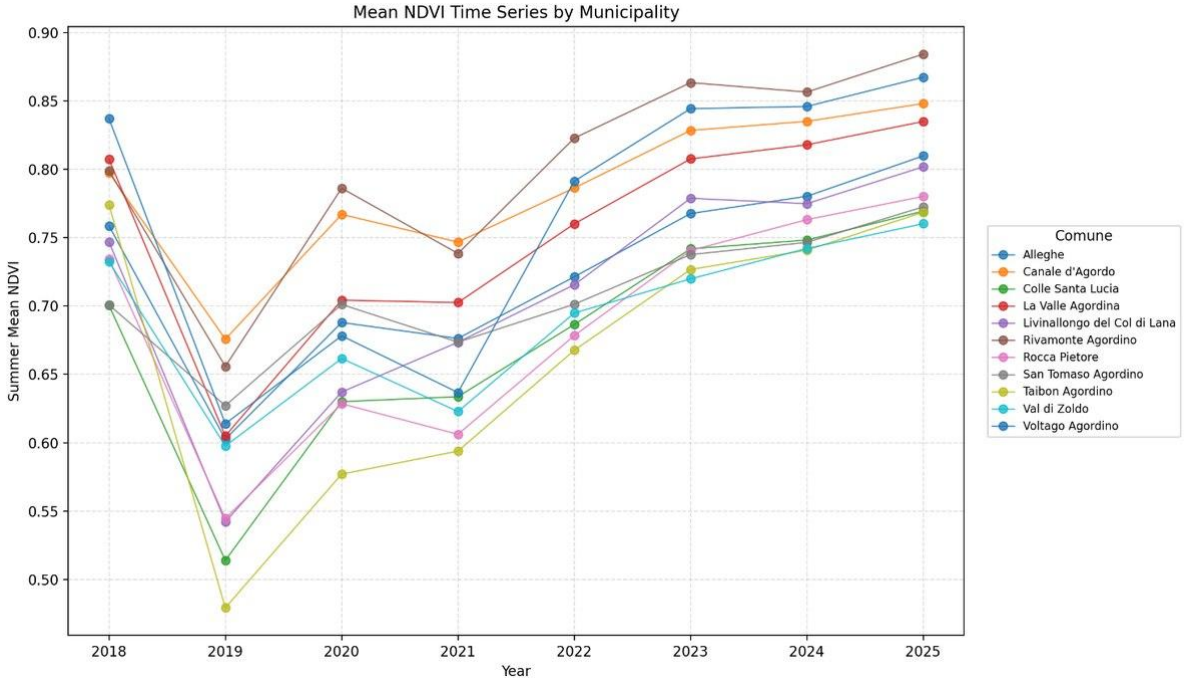


Figure 6: Annual July–August mean NDVI (2018–2025) by municipality within the AOI.

7. Discussion

NDVI proved effective for damage assessment. As Section 6 shows, differencing pre and post storm summer composites (ΔNDVI, Figure 3) correctly identified the area of most severe canopy loss, and these closely coincided with the official damage perimeters declared by Regione Veneto (Figure 4)

The outcome is different when the same index is asked to track recovery over time: NDVI suggests that vegetation cover in most municipalities has returned to, or exceeded, pre-storm (2018) summer levels by 2025 - yet visual inspection of recent satellite basemaps over the same stands shows the forest canopy has clearly not regrown to its pre-storm state. This is not an error in the data but a limitation of NDVI itself: the index responds to the photosynthetic greenness of whatever vegetation is present at the surface - including the grasses, ferns and shrubs that colonise a wind-thrown slope quickly, and which can dominate the NDVI signal well before a multi-layered conifer canopy has had time to re-establish (Emmett et al., 2019). A dense summer meadow of herbaceous regrowth and a mature spruce–fir stand can therefore produce very similar NDVI values. This mirrors an independent study of two other Vaia-damaged spruce stands in the Southern Alps, which likewise found NDVI on track to reach pre-storm levels within about a decade while concluding that spontaneous restoration of the original mature forest community was unlikely over the same timeframe (Giupponi et al., 2023). In short: useful for locating where the damage was worst, unreliable for confirming that it has healed.

A more truthful picture of recovery would require indices and data sources that respond to canopy structure and biomass, not only to surface greenness. Four additions would directly address the discrepancy observed here:

- NBR (Normalized Burn Ratio), using the SWIR2 band is less prone to early saturation than NDVI; boreal-forest studies using Landsat time series found NDVI tracks mainly the initial flush of herbaceous vegetation, whereas NBR responds more to the increasing canopy complexity of tree re-establishment (Pickell et al., 2016) - it would likely still separate structurally recovered stands from still-damaged ones where NDVI has already plateaued.
- EVI corrects for canopy background reflectance and remains sensitive to canopy variation in high-biomass conditions where NDVI has been shown to saturate (Huete et al., 2002), so it should track canopy development further into the recovery process instead of flattening out early.
- NDMI, built on the same NIR/SWIR canopy-water-content principle validated by Gao (1996), differs meaningfully between a closed, shaded mature canopy and an open herbaceous or shrub layer, which could help separate the two even when their NDVI values are similar.
- Sentinel-1 SAR backscatter - specifically the Radar Vegetation Index (RVI), a dual-polarisation VV/VH formulation - can give additional information: radar response is driven by woody volume and canopy structure rather than chlorophyll, so a grass- or shrub-covered slope would show markedly lower backscatter than an intact mature stand even when both look equally “green” in NDVI. Combining SAR with optical data has been shown to significantly reduce exactly this kind of saturation in high-biomass forest (David et al., 2022). Of the four, this is identified as the single most informative addition for resolving the discrepancy observed here - and being weather- and illumination-independent, it is particularly valuable in a cloud-prone alpine environment.

Google Earth Engine enabled efficient, server-side processing of the full multi-temporal Sentinel-2 archive over the study period without needing to download or locally store the underlying imagery -

a workflow pattern well suited to incorporating exactly these additional bands and sensors at larger scale.

8. Limitations

- This analysis relies on a single index, NDVI, which cannot reliably distinguish surface-level greenness from true canopy structural recovery and on its own tends to overstate how far a forest has actually recovered.
- The CORINE 2018 forest baseline has a 100 m pixel size and a 25-hectare minimum mapping unit, considerably coarser than the 10–20 m Sentinel-2 data used elsewhere in the analysis; small or narrow forest patches may be under-represented in the mask.
- Validation relied on qualitative visual cross-referencing against a disturbance layer rather than a field-based accuracy assessment.

9. Conclusions & Future Work

This project demonstrates that a transparent, reproducible cloud-based workflow - built entirely on free and open satellite data - can meaningfully identify damaged areas.

A first, low-effort refinement would be to test an earlier-season compositing window - closer to June than the July–August window used throughout this analysis - since a comparable subalpine conifer forest study found NDVI–recovery correlations to be more reliable earlier in the growing season, before herbaceous and forb cover peaks and can overwhelm the structural recovery signal (Buma, 2012). Unlike the extensions below, this would need no new data sources, only reprocessing the existing Sentinel-2 archive over a different month range.

The results of this study show a workflow that is effective at one task and appropriately cautious about another. As a means of locating storm damage, the ΔNDVI approach performed well, closely matching Regione Veneto's own official damage perimeters. Recovery monitoring, however will need more than NDVI alone. A natural next step would be to test whether the four indices identified in Section 7 - NBR, EVI, NDMI and RVI - could together support the construction of a forest resilience index, and if so, which combination performs best.

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